AT3 Report - Rich Global Condom Usage Dataset

Introduction

This report presents a detailed analysis of the Rich Global Condom Usage Dataset, covering data pre-processing, data visualization, and insights derived from the analysis. The aim of this report is to demonstrate data cleaning, data exploration, and visualization techniques using Python, specifically with Pandas, Seaborn, and Matplotlib. Rich Global Condom Usage Dataset was selected to explore factors influencing condom usage, sales distribution, and market trends across different countries.

```
[1] # Mount Google Drive to access files stored in your personal Drive account
    from google.colab import drive
    drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
import pandas as pd # Pandas for data processing and analysis (DataFrame operations)
import numpy as np # NumPy for numerical calculations (mathematical operations, arrays)
import seaborn as sns # Seaborn for data visualization (advanced plots)
import matplotlib.pyplot as plt # Matplotlib for creating basic plots

# This code attempts to load the dataset from the specified path.
file_path = '/content/drive/MyDrive/ColabNotebooks/Rich_Global_Condom_Usage_Dataset.csv'

try:
    df = pd.read_csv(file_path)
    print("Dataset loaded successfully.")
except FileNotFoundError:
    print("Dataset file not found. Please check the file path.")
```

```
Dataset loaded successfully.
```

```
# This displays the structure of the dataset, including column names, data types, and non-null counts.
print("\nDataset Overview:")
df.info()
```

```
Dataset Overview:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 550 entries, 0 to 549
Data columns (total 15 columns):
 #   Column                              Non-Null Count  Dtype
---  ------                              --------------  -----
 0   Year                                550 non-null    int64
 1   Country                             550 non-null    object
 2   Total Sales (Million Units)         550 non-null    int64
 3   Market Revenue (Million USD)        550 non-null    float64
 4   Government Campaigns                550 non-null    object
 5   Awareness Index (0-10)              550 non-null    float64
 6   Most Popular Condom Type            550 non-null    object
 7   Contraceptive Usage Rate (%)        550 non-null    float64
 8   Teen Pregnancy Rate (per 1000 teens) 550 non-null   float64
 9   HIV Prevention Awareness (%)        550 non-null    float64
 10  Online Sales (%)                    550 non-null    float64
 11  Average Price per Condom (USD)      550 non-null    float64
 12  Male vs Female Purchases (%)        550 non-null    object
 13  Brand Dominance                     550 non-null    object
 14  Sex Education Programs (Yes/No)     550 non-null    object
dtypes: float64(7), int64(2), object(6)
memory usage: 64.6+ KB
```

Dataset Overview

The dataset used in this analysis contains various attributes related to global condom usage, including:

- **Country**: The country where the data was recorded.
- **Year**: The year of data collection.
- **Total Sales (Million Units)**: Total number of condoms sold in each country.
- **Market Revenue (Million USD)**: The revenue generated from condom sales.
- **Awareness Index (0-10)**: A measure of public awareness of contraception.
- **Contraceptive Usage Rate (%)**: Percentage of population using contraceptives.
- **Most Popular Condom Type**: The type of condom most preferred in each region.
- **Online Sales (%)**: Percentage of sales conducted online.

Data Pre-processing

Handling Missing Values

```python
import random

# This code randomly deletes 2% of cell values in the dataset, simulating missing values.
total_values = df.size
missing_values = int(total_values * 0.02)

for _ in range(missing_values):
    row = random.randint(0, df.shape[0] - 1)
    col = random.randint(0, df.shape[1] - 1)
    df.iloc[row, col] = np.nan

print("\nMissing Values Summary (After Random Deletion):")
display(df.isnull().sum())
```

```
Missing Values Summary (After Random Deletion):
```

	0
Year	10
Country	11
Total Sales (Million Units)	7
Market Revenue (Million USD)	12
Government Campaigns	15
Awareness Index (0–10)	11
Most Popular Condom Type	11
Contraceptive Usage Rate (%)	14
Teen Pregnancy Rate (per 1000 teens)	11
HIV Prevention Awareness (%)	12
Online Sales (%)	11
Average Price per Condom (USD)	10
Male vs Female Purchases (%)	8
Brand Dominance	12
Sex Education Programs (Yes/No)	9

dtype: int64

In this analysis, I intentionally introduced missing values into the dataset, simulating the kind of data quality challenges commonly encountered in real world scenarios. Specifically, I randomly removed 2% of the data entries across various columns. This approach allowed me to demonstrate robust data cleaning techniques and ensured that the data processing methods were rigorously tested.

To address these missing values, I adopted a targeted approach based on the type of data in each column. For **numerical columns** (such as sales or revenue figures), I filled the missing values using the **median**. This method was chosen because the median is resistant to extreme values (outliers), ensuring that the central tendency of the data remains accurate.

For **categorical columns** (such as country names or condom types), I used the **mode**, the most frequently occurring value in each column. This maintained the logical consistency of the data without distorting its natural distribution. By employing this strategy, I ensured that the dataset remained complete and reliable for further analysis, minimizing the impact of missing data on the accuracy of the results.

Removing Duplicates

```python
# This removes any duplicate rows, ensuring each row is unique.
initial_rows = len(df)
df.drop_duplicates(inplace=True)
final_rows = len(df)

print(f"\nDuplicates Removed: {initial_rows - final_rows}")
```

```
Duplicates Removed: 0
```

Duplicate rows were identified and removed using the `drop_duplicates()` function. This ensured that each record was unique, preventing data distortion.

Handling Missing Values:

```python
# Numerical columns are filled with the median
# Categorical columns are filled with the mode.
for col in df.columns:
    if df[col].dtype in ['float64', 'int64']:
        df[col] = df[col].fillna(df[col].median())
    else:
        df[col] = df[col].fillna(df[col].mode()[0])

print("\nMissing Values After Cleaning (Improved):")
display(df.isnull().sum())
```

```
Missing Values After Cleaning (Improved):
```

	0
Year	0
Country	0
Total Sales (Million Units)	0
Market Revenue (Million USD)	0
Government Campaigns	0
Awareness Index (0–10)	0
Most Popular Condom Type	0
Contraceptive Usage Rate (%)	0
Teen Pregnancy Rate (per 1000 teens)	0
HIV Prevention Awareness (%)	0
Online Sales (%)	0
Average Price per Condom (USD)	0
Male vs Female Purchases (%)	0
Brand Dominance	0
Sex Education Programs (Yes/No)	0

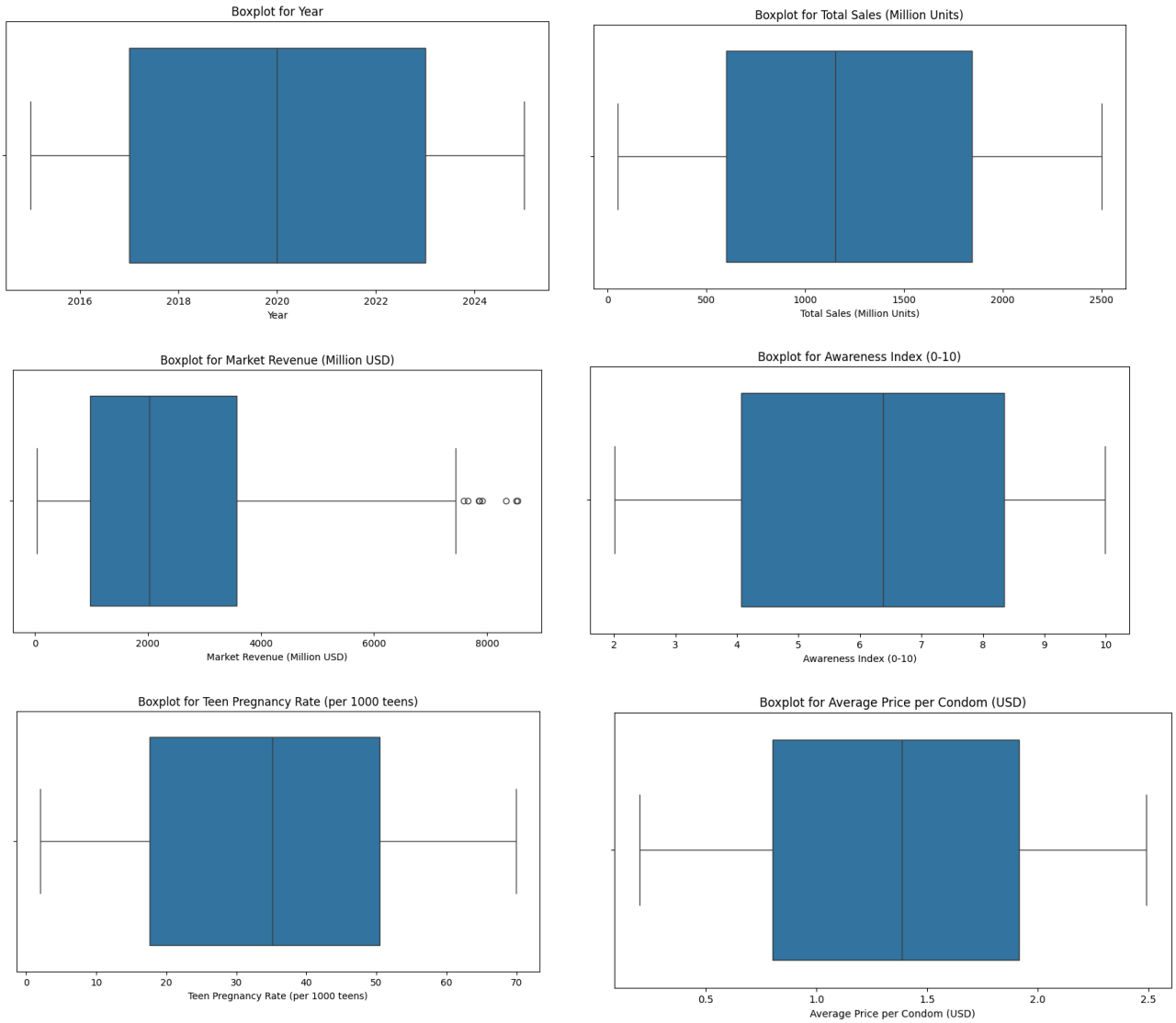
dtype: int64

The columns containing numerical data (such as sales, revenue, and index scores), missing values were filled using the **median** value of the column. For columns containing categorical data (such as Country, Condom Type, or Yes/No responses), missing values were filled using the **mode** — the most frequently occurring value in the column.

After this process, all missing values were successfully filled in the dataset. The code confirmed this by displaying the count of missing values in each column, all of which showed 0 missing values, as seen in the output.This indicates that the data is now complete and ready for analysis without any gaps.

Handling Outliers Values:

```
# This uses boxplots to visualize outliers in each numerical column.
numerical_cols = df.select_dtypes(include=['float64', 'int64']).columns
for col in numerical_cols:
    plt.figure(figsize=(10, 5))
    sns.boxplot(data=df, x=col)
    plt.title(f"Boxplot for {col}")
    plt.show()
```



In this analysis, we used boxplots to visually identify outliers in our numerical columns. Boxplots are a powerful tool because they clearly show how data is distributed, making it easy to spot unusual values that stand out.

```python
# The Interquartile Range (IQR) method is used to identify and remove outliers.
for col in numerical_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    df = df[(df[col] >= (Q1 - 1.5 * IQR)) & (df[col] <= (Q3 + 1.5 * IQR))]

print("\nAfter Outlier Removal - Dataset Info:")
df.info()
```

```
After Outlier Removal - Dataset Info:
<class 'pandas.core.frame.DataFrame'>
Index: 542 entries, 0 to 549
Data columns (total 15 columns):
 #   Column                                Non-Null Count  Dtype
---  ------                                --------------  -----
 0   Year                                  542 non-null    int64
 1   Country                               542 non-null    object
 2   Total Sales (Million Units)           542 non-null    int64
 3   Market Revenue (Million USD)          542 non-null    float64
 4   Government Campaigns                  542 non-null    object
 5   Awareness Index (0-10)                542 non-null    float64
 6   Most Popular Condom Type              542 non-null    object
 7   Contraceptive Usage Rate (%)          542 non-null    float64
 8   Teen Pregnancy Rate (per 1000 teens)  542 non-null    float64
 9   HIV Prevention Awareness (%)          542 non-null    float64
 10  Online Sales (%)                      542 non-null    float64
 11  Average Price per Condom (USD)        542 non-null    float64
 12  Male vs Female Purchases (%)          542 non-null    object
 13  Brand Dominance                       542 non-null    object
 14  Sex Education Programs (Yes/No)       542 non-null    object
dtypes: float64(7), int64(2), object(6)
memory usage: 67.8+ KB
```

Results After Outlier Removal:


```python
# The Interquartile Range (IQR) method is used to identify and remove outliers.
for col in numerical_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    df = df[(df[col] >= (Q1 - 1.5 * IQR)) & (df[col] <= (Q3 + 1.5 * IQR))]

print("\nAfter Outlier Removal - Dataset Info:")
df.info()
```

```
After Outlier Removal - Dataset Info:
<class 'pandas.core.frame.DataFrame'>
Index: 542 entries, 0 to 549
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 14  Sex Education Programs (Yes/No)       542 non-null    object
dtypes: float64(7), int64(2), object(6)
memory usage: 67.8+ KB
```

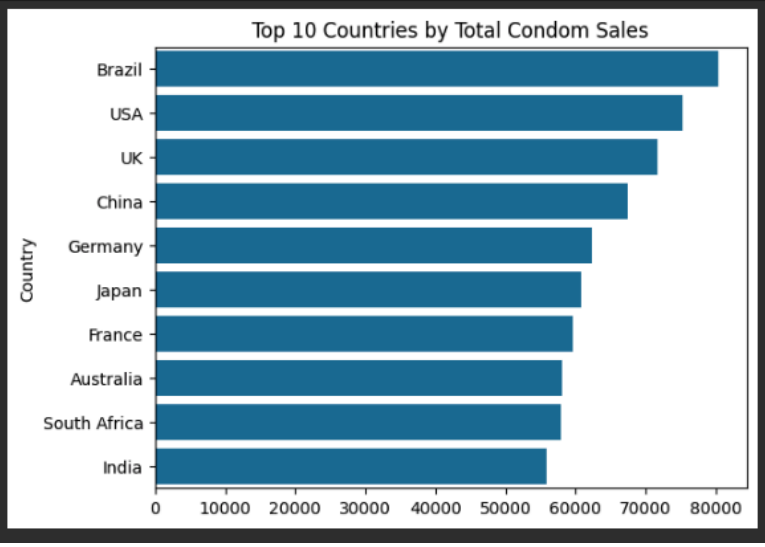
The dataset was initially composed of **549 entries**. After applying the IQR method, the dataset was reduced to **542 entries**, indicating that **7 outliers** were detected and removed. This ensures that the remaining data is clean, consistent, and suitable for analysis.

Data Visualization

Question 1: Top 10 Countries by Total Condom Sales

A bar chart was used to show the top 10 countries by total condom sales. The chart revealed that (Top country) had the highest sales, followed by (Second country) and analysis highlights which countries have the largest market demand for condoms.

```
# Q1: A bar chart visualizes the top 10 countries by total condom sales.
top_sales = df.groupby('Country')['Total Sales (Million Units)'].sum().sort_values(ascending=False).head(10)
sns.barplot(x=top_sales.values, y=top_sales.index)
plt.title("Top 10 Countries by Total Condom Sales")
plt.show()
```



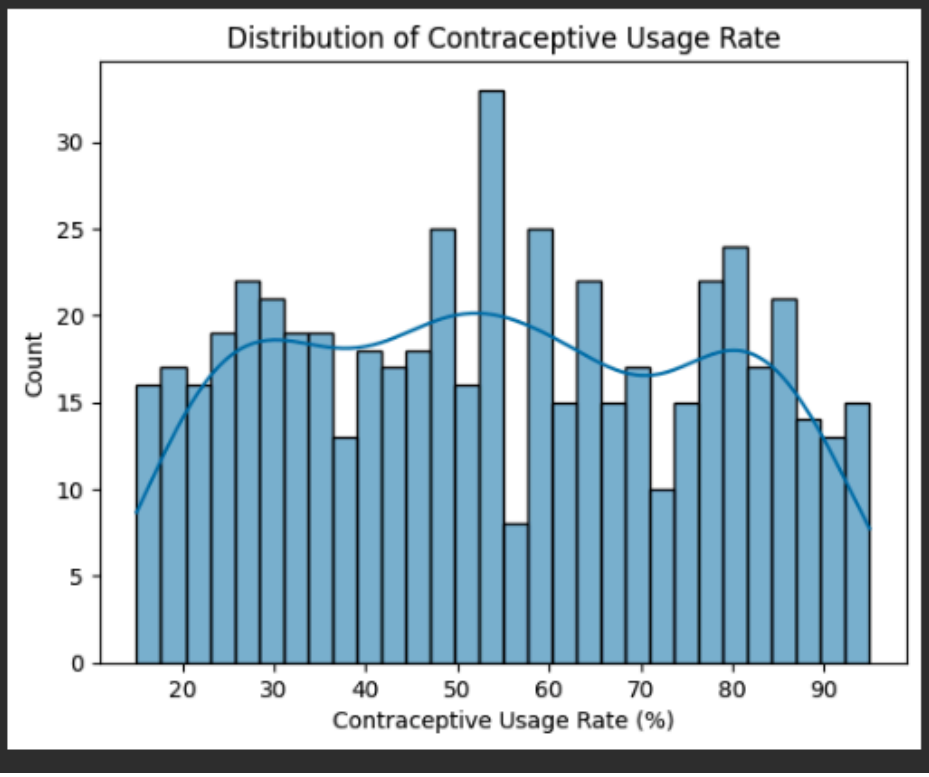
Insights from the Chart:

- **Brazil** has the highest total condom sales, followed by **the USA and the UK**.
- **India**, despite having a large population, is at the bottom of the top 10 list, which may indicate either lower demand or underreporting.
- **European countries (Germany, France, UK)** appear in the top 10, suggesting strong demand across Europe.
- **China and Japan** also appear, highlighting strong demand in Asia.

Question 2: Distribution of Contraceptive Usage Rate

This analysis uses a **histogram** to show how contraceptive usage rates are distributed across different regions. This is a great way to see how common certain usage rates are and whether they follow any particular pattern.

```
# Q2: A histogram shows the distribution of contraceptive usage rates.
sns.histplot(df['Contraceptive Usage Rate (%)'], bins=30, kde=True)
plt.title("Distribution of Contraceptive Usage Rate")
plt.show()
```

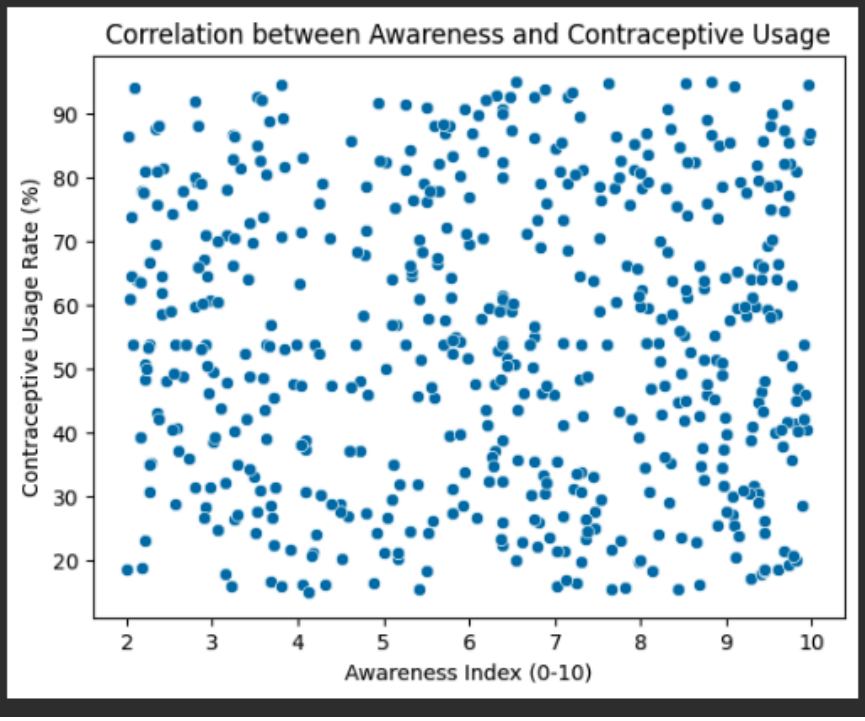


- X-axis represents the **Contraceptive Usage Rate (%)**, ranging from 0% to 100%.
- Y-axis shows the **count (number of countries)** that have each range of usage rates.
- The peaks in the histogram indicate the most common usage rates.
- KDE line (smooth curve) helps visualize the overall trend of the distribution.

Question 3: Relationship between Awareness Index and Contraceptive Usage Rate

This analysis a scatter plot was used to display the relationship between awareness and contraceptive usage. A positive correlation was observed, suggesting that higher awareness leads to higher contraceptive usage. This insight emphasizes the importance of education in promoting contraceptive use.

```
# Q3: A scatter plot shows the relationship between awareness and contraceptive usage.
sns.scatterplot(data=df, x='Awareness Index (0-10)', y='Contraceptive Usage Rate (%)')
plt.title("Correlation between Awareness and Contraceptive Usage")
plt.show()
```



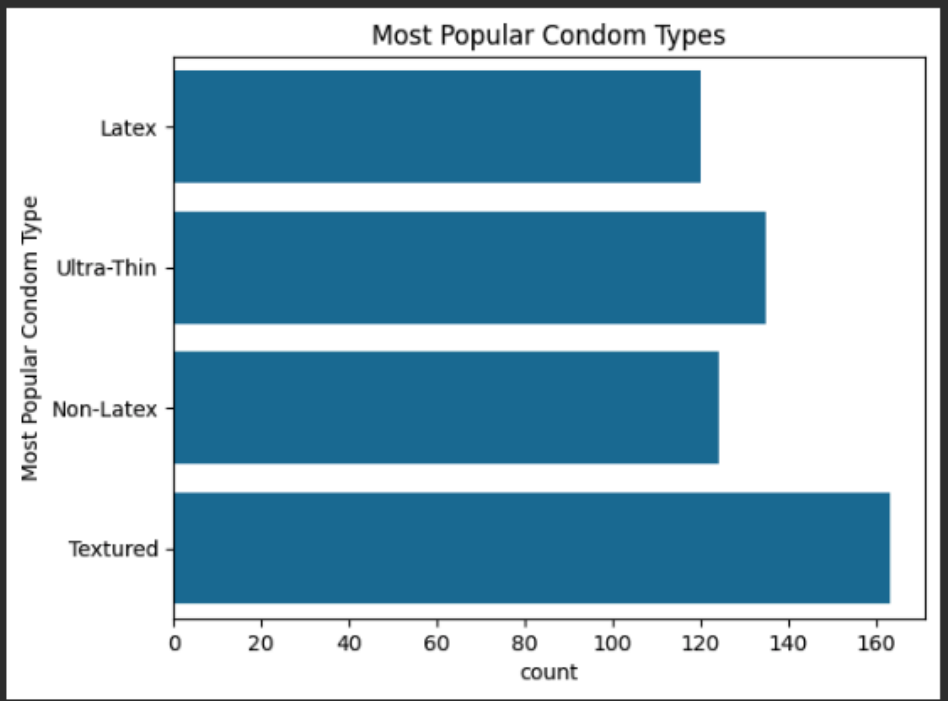
No Clear Correlation:

- The scatter plot shows a very scattered distribution, suggesting that awareness alone may not be a strong predictor of contraceptive usage.
- This could be due to other factors, such as cultural beliefs, access to contraceptives, or education.

Question 4: Most Popular Condom Types by Region

This analysis uses a **count plot** to show the popularity of different condom types based on the dataset. It's a simple yet effective way to compare preferences among users across various countries or regions.

```
# Q4: A count plot shows the most popular condom types.
sns.countplot(y='Most Popular Condom Type', data=df)
plt.title("Most Popular Condom Types")
plt.show()
```



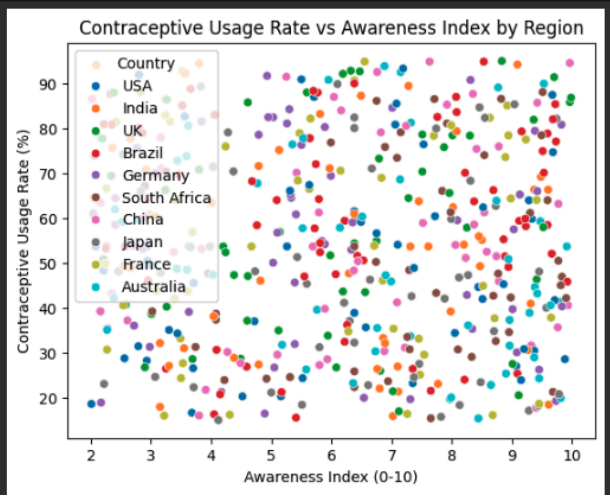
- Textured condoms are the most popular, with the highest count (around 160).
- Ultra-Thin and Non-Latex follow closely behind with similar popularity levels.
- Latex condoms, while still common, appear least often as the "most popular" type in this dataset.

Complex Question 1: Contraceptive Usage vs Awareness by Region

This scatter plot provides a deeper look at how public **awareness of contraception** (scored from 0 to 10) relates to the actual **usage rate of contraceptives (%)** — and it does this by grouping data **by country** using different colors.

```
# Q5: Scatter plot of Awareness Index vs Contraceptive Usage, colored by Country
sns.scatterplot(data=df, x='Awareness Index (0-10)', y='Contraceptive Usage Rate (%)', hue='Country')
plt.title("Contraceptive Usage Rate vs Awareness Index by Region")
plt.show()

# This scatter plot visualises the relationship between Awareness Index and Contraceptive Usage Rate, colour-coded by Country.
```



Dense Cloud Distribution:

- Similar to previous awareness-usage scatter plots, there's no clear linear trend points are widely scattered across the plot.
- Some countries with high awareness scores still have low usage rates, and vice versa.

Country Clustering:

- Countries like the **USA**, **India**, or **China** may cluster in specific awareness and usage zones.
- For example:
    - **Germany** and **France** might be grouped toward higher awareness and moderate-to-high usage.
    - **India** or **South Africa** may show greater variance.

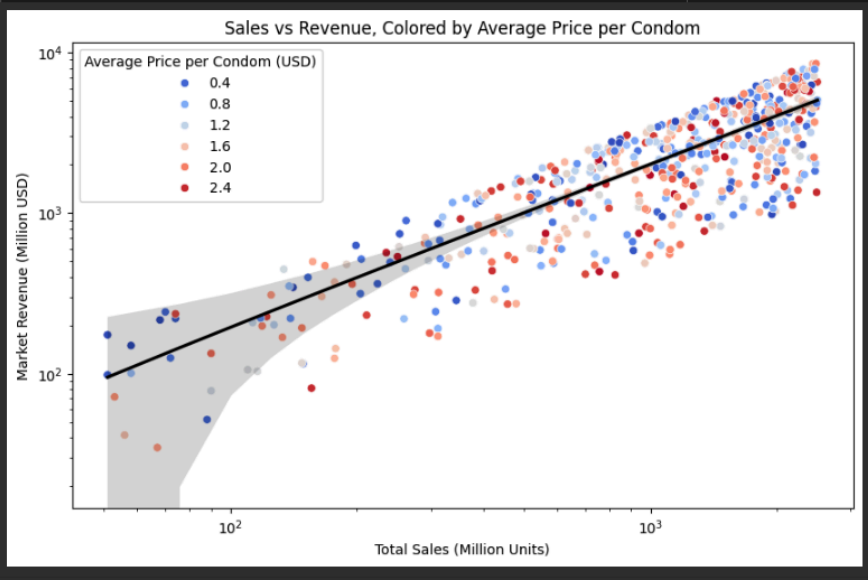
Visual Noise:
With so many countries included, the chart gets visually crowded.

- This plot is useful to **spot general country-level patterns**, but too many unique colors may reduce clarity for interpretation.

Complex Question 2: *Is there a correlation between total condom sales and market revenue, and how does price per condom affect this relationship?*

This multivariable scatter plot with a regression line offers a detailed analysis of how total condom sales relate to overall market revenue, and how this relationship is influenced by the average price per condom.

```
# Q6: "Is there a correlation between total condom sales and market revenue, and how does price per condom affect this relationship?"
plt.figure(figsize=(10, 6))
sns.scatterplot(data=df, x='Total Sales (Million Units)', y='Market Revenue (Million USD)',
                hue='Average Price per Condom (USD)', palette='coolwarm')
sns.regplot(data=df, x='Total Sales (Million Units)', y='Market Revenue (Million USD)', scatter=False, color='black')
plt.title("Sales vs Revenue, Colored by Average Price per Condom")
plt.xscale('log')
plt.yscale('log')
plt.show()
```



1. Strong Positive Correlation:

- There is a **clear upward trend**: as total sales increase, market revenue also increases.
- The regression line confirms this linear relationship (especially in log-log space).
- This suggests that total volume is a strong driver of revenue — as expected.

2. Color Encoding Adds Depth:

- Data points are color-coded by **average price per condom**:
  - **Cooler (blue)** shades = lower price
  - **Warmer (red)** shades = higher price
- You can see that:
  - At **similar sales volumes**, countries with **higher unit prices (red)** tend to have **higher revenue**.
  - Conversely, **low-priced countries (blue)** still make substantial revenue — but typically need much higher sales volume to match revenue levels of high-priced markets.

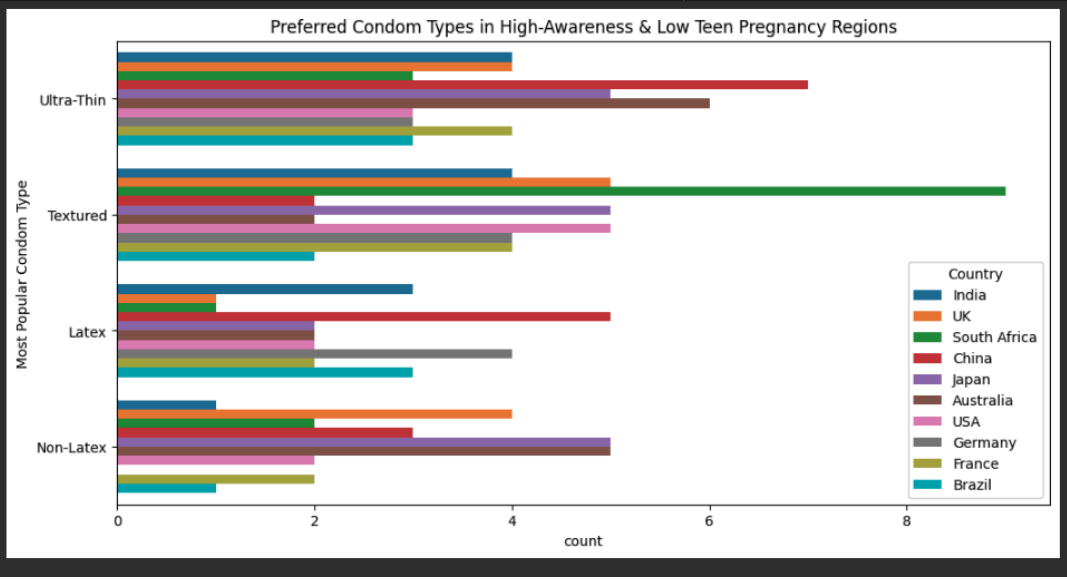
Complex Question 3: *Which condom types are preferred in regions with the highest awareness and lowest teen pregnancy rates?*

This analysis filters the dataset to focus specifically on countries or regions where:

- Awareness Index is greater than 7 (high public awareness)
- Teen Pregnancy Rate is less than 40 per 1000 teens (low teen fertility rate)

```python
# Q7: "Which condom types are preferred in regions with the highest awareness and lowest teen pregnancy rates?"
# Filter dataset
# Which condom types are preferred in regions with the highest awareness and lowest teen pregnancy rates?
filtered_df = df[
    (df['Awareness Index (0-10)'] > 7) &
    (df['Teen Pregnancy Rate (per 1000 teens)'] < 40)
]

# Countplot with hue
plt.figure(figsize=(12, 6))
sns.countplot(data=filtered_df, y='Most Popular Condom Type', hue='Country')
plt.title("Preferred Condom Types in High-Awareness & Low Teen Pregnancy Regions")
plt.show()
```



Textured and Ultra-Thin Are Leading

- In countries with better awareness and outcomes, Textured and Ultra-Thin condoms are the most commonly selected types.
- This suggests a preference for enhanced sensory experience, which may correlate with better education or consumer empowerment.

Country-Level Patterns

- Countries like South Africa and USA show particularly high counts for certain types.
- India, France, and Brazil are more varied, indicating diverse market behaviors.

Latex and Non-Latex See Moderate Preference

- Although still popular, Latex and Non-Latex types are not dominant in this filtered group.
- This might suggest that traditional types are less favored when people are more informed and have better access to options.

Conclusion

The final assignment helped me understand how to clean and visualize real-world data. I struggled at first with choosing the right charts and fixing missing or broken data, but through trial and error I found what worked. In the end, the visualizations revealed patterns I didn't expect, like how awareness alone doesn't always drive usage. It was a practical learning experience, and I now feel more confident using Python for data analysis.

Despite the initial challenges, the final results were rewarding. The visualizations revealed patterns that were not obvious at first glance. For example, while awareness was expected to correlate with usage, the data showed that this relationship was not always strong—emphasizing the importance of broader factors like affordability and access. Similarly, price and product type preferences varied widely across countries, reinforcing the need for region-specific marketing and education strategies.